EPFL — Fall 2026

COM-309

Introduction to quantum information processing

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This document was originally created for my own use, but I am sharing it because I believe it may be useful to others for studying and reference purposes. However mistake can happen and there will inevitably be some errors, if you found any, you are more than welcome to send me a message (Arthur Hbe on telegram) or to directly do a PR on the repo

`https://github.com/Arthur926564/LectureNotes`

If you think that something is missing or that something is not clear, I will be more than happy to add your modification to this document.

Unless otherwise stated, all material in these notes is based on the course content, including the diagrams, explanations, and examples.

Note also that these notes were taken during the lecture, but not only; some materials was added later (that's why you may read remarks that sounds like I knew the course before taking it).

Finally, thanks a lot to Joachim Favre for his help with the setup and whose style is used in this document.

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Introduction

Overview of the course

Quantum information processing is a sort of unification of the two followings fields: **physics** and **computer science**/ **communication**. Quantum physics is a new field of physics that emerged around 1900/1930, we found that that the behaviour of particle at a really small scale was really different of what we expected, uncertainty was the new discovery at atomic scale. We are not able to exactly predict how a particle behave even by knowing every information that could possibly be known.

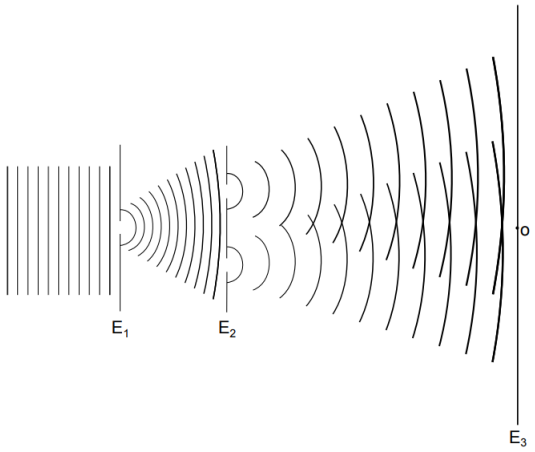
Documents

The class materials are typed into lecture notes which covers this course and the quantum computation course CS-308 (chapters 9, 10, 11, 12, 13, 14).

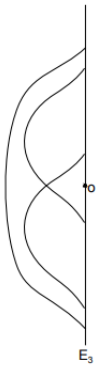
Chapter 1

Wave-Particle Duality

Interference in Double slit exp.



What this has shown is that light behaved like a wave in our case. If light were in fact particle, the result of the following experiences would be different. The resulting wave would be something like this:



However lights behaving like a wave is something that we can all agree on, on the other case, imagine doing the same experiment but for electrons or C60, the resulting frequency would corresponds to a wave as for lights. And this frequency is **constant** i.e., we can send electrons one after the other then after collecting the statistics; remarkably the function we get is the same prediction as the one of wave.

Chapter 2

Mathematical notions

2.0.1 Recap of linear algebra in Dirac notation

The arena for quantum mechanics is Hilbert space (a vector space on $\mathbb{C}$).

Notation

The language in order to speak in quantum science is **Dirac** notation.

Definition 1. Ket

$$\vec{\varphi} = |\varphi\rangle = \begin{pmatrix} \varphi_1 \\ \varphi_1 \\ \vdots \\ \varphi_n \end{pmatrix} \quad , \varphi_i \in \mathbb{C}$$

Definition 2. Bra

$$\vec{\varphi}^\dagger = \langle\varphi| = (\varphi_i^*, \varphi_2^*, \ldots, \varphi_D^*), \varphi_i \in \mathbb{C}$$

Kets and Bras are the foundation of the language of quantum science, those will be used extensively throughout the whole course.

Definition 3. Inner product In our new space, we will define the inner product as

$$\vec{\varphi}^\dagger \cdot \vec{\varphi} = \sum_{i=1}^{D} \varphi_i^* \cdot \varphi_i$$

Properties

As for vector in classical vector space, vector inside Hilbert space also end up having properties that we can use in order to prove theorems:

$$\langle\varphi||\varphi\rangle^* = \langle\varphi||\varphi\rangle$$
$$(\alpha^*\langle\varphi_1| + \beta^*\langle\varphi_2|)|\varphi\rangle = \alpha^*\langle\varphi_1||\varphi\rangle + \beta^*\langle\varphi_2||\varphi\rangle$$

The positivity of bracket:

$$\langle\varphi||\varphi\rangle \geq 0$$

We also have triangle inequality and Cauchy-Schwartz:

$$\|\varphi + \varphi\| \leq \|\varphi\| + \|\varphi\|$$
$$|\langle\varphi||\varphi\rangle| \leq \|\varphi\|\|\varphi\|$$

Orthonormal basis in $\mathcal{H}$

the dimension of our space is denoted as $\dim\mathcal{H} = D$. The basis of that space is composed of basis vector which are noted as:

$$|u_1\rangle, |u_2\rangle, \ldots, |u_D\rangle$$

Where

$$\langle u_i||u_j\rangle = \begin{cases} 1 & i = j \\ 0 & j \neq i \end{cases}$$

as vector that are not equal are orthogonal hence their inner product equals 0.

Example The most commons space that will be used during this course is $\mathbb{C}^2 = \left\{ \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \alpha, \beta \in \mathbb{C} \right\}$. In this case we define our vector as

$$|\varphi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Tensor product

The goal of a tensor product is to combine spaces together.

Definition 4. Given $\mathcal{H}_1$ and $\mathcal{H}_2$ we define $\mathcal{H}_1 \otimes \mathcal{H}_2$ as Where $i = 1, \ldots, D_1$ and $j = 1, \ldots, D_2$.

Example Taking again $\mathbb{C}^2$ We want to convert it into $\mathbb{C}^2 \otimes \mathbb{C}^2$,the new basis of our space this gives:

$$\begin{aligned} |0\rangle \otimes |0\rangle &= |0\ 0\rangle \\ |0\rangle \otimes |1\rangle &= |0\ 1\rangle \\ |1\rangle \otimes |0\rangle &= |1\ 0\rangle \\ |1\rangle \otimes |1\rangle &= |1\ 1\rangle \end{aligned}$$

This allows us to have multiple qbits inside one state, which later on will allow us to have the notion of entanglement.

If we wanted to translate this into classical linear algebra, this result for the tensor product would be the following: given two vectors $\vec{v} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ and $\vec{u} = \begin{pmatrix} \gamma \\ \delta \end{pmatrix}$ the tensor product would be computed as

$$\vec{v} \otimes \vec{u} = \begin{pmatrix} \alpha \begin{pmatrix} \gamma \\ \delta \end{pmatrix} \\ \beta \begin{pmatrix} \gamma \\ \delta \end{pmatrix} \end{pmatrix} = \begin{pmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{pmatrix}$$

Inner product in $\mathcal{H}_1 \otimes \mathcal{H}_2$

$$(\langle\varphi_1| \otimes \langle\varphi_2|)(|\varphi_1\rangle \otimes |\varphi_2\rangle) = \langle\varphi_1||\varphi_1\rangle\langle\varphi_2||\varphi_2\rangle$$

Matrix

Definition 5. A matrix $A : \mathcal{H} \to \mathcal{H}$ is an operator defined in a given an orthonormal basis $\{|1\rangle, |2\rangle, \ldots, |D\rangle\}$ which can be represented as an $D \times D$ array where $A_{ij} = \langle i|A|j\rangle$. We can even define our matrix as:

$$A = \sum_{i,j=1}^{D} A_{ij}|i\rangle\langle j|$$

Where $|i\rangle\langle j|$ is the matrix that has a 1 at the position (i, j) and 0 everywhere else.

Tensor product of matrices Given two Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$ we know how to compute the inner product space of the tensor product between our two spaces

$\mathcal{H}_1, \mathcal{H}_2$ as seen before. What we want to see now is the tensor product between two matrices A and B.

To do so we'll use the definition of the matrix using the '*ket bra*' notation of the definition above.

$$A \otimes B = \sum_{i,j,j,l} A_{ij} B_{jl} \underbrace{|i\rangle\langle j| \otimes |k\rangle\langle l|}_{(|i\rangle \otimes |k\rangle)(\langle j| \otimes \langle l|)}$$

Remark 1. The notation $|i\rangle \otimes |k\rangle$ can be simplified by $|i, k\rangle$.

$$A \otimes B = \sum_{i,j=1}^{D_1} \sum_{k,l}^{D_2} A_{ijkl} |i, k\rangle\langle j, l|$$

The resulting dimension of our matrix is given by $D_1 D_2 \times D_1 D_2$ where

$$(A \otimes B)_{ik,jl} = A_{ij} B_{kl}$$

Example

Given the product of $\mathbb{C}^2 \otimes \mathbb{C}^2$, let us see what would be the tensor product of the two matrices:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$$

The goal is to compute $A \otimes B$.

$$\begin{aligned} A \otimes B &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \otimes \begin{pmatrix} e & f \\ g & h \end{pmatrix} \\ &= \begin{pmatrix} a \cdot B & bB \\ cB & dB \end{pmatrix} \\ &= \begin{pmatrix} ae & af & be & bf \\ ad & ah & bd & bh \\ ce & cf & ge & gf \\ cd & ch & gd & gh \end{pmatrix} \end{aligned}$$

2.0.2 Hermitian matrix

First let us do some notation:

$$\begin{aligned} A^\dagger &= \sum_{i,j} A_{ij}^* |j\rangle\langle i| \\ &= \sum_{i,j} A_{ij}^* |i\rangle\langle j| \end{aligned}$$

We have the following:

$$(A^*)_{ij} = A_{ij}^*$$

Hermitiam matrices

Definition 6. An Hermitian matrix is a given operator A on an Hilbert space $\mathcal{H}$ where

$$A = A^\dagger$$

Unitary matrices

Definition 7. An unitary matrix is an Hermitian operator U on an Hilbert space $\mathcal{H}$ where

$$U^{-1} = U^{\dagger}$$

Here an equivalent property can be derived

$$UU^{\dagger} = U^{\dagger}U = I$$

Which is equivalent to say for two any given pure states $|\varphi\rangle, |\varphi\rangle$

$$\langle\varphi||\varphi\rangle = \langle\varphi|UU^{\dagger}|\varphi\rangle = (U|\varphi\rangle)^{\dagger} \cdot (U|\varphi\rangle)$$

Spectral theorem for Hermitian matrices

Theorem 1. Given a Hermitian matrices A we have that all eigenvalues of our matrix are real number where their respective eigenvector are orthonormal.

$$A|u_i\rangle = \lambda_i|u_i\rangle$$

with $\lambda_i \in \mathbb{R}$.

In particular, we have:

$$A = \sum_{i=1}^{d} \lambda_i |v_i\rangle\langle v_i|$$

Chapter 3

Quantum Physics

3.1 Principles of Quantum Physics

In this section, we will introduce the principles in an axiomatic form. We will formulate them in algebraic manner appropriate for quantum information and computation in finite dimensional Hilbert space.

Hilbert space

Note here that we have not defined what a Hilbert space here is so let me define it (wikipedia)

Definition 8. (Hilbert Space) A **Hilbert** space is a real or complex inner product space that is also a complete metric space with respect to the distance function induced by the inner product

To say that a complex vector space $\mathcal{H}$ is a complex inner product space means that there is an inner product $\langle x, y\rangle$ associating a complex number to each pair of elements x, y of $\mathcal{H}$ that satisfies the following properties:

1. The inner product is conjugate symmetric; that is the inner product of a pair of elements is equal to the complex conjugate of the inner product of the swapped elements:

$$\langle y, x\rangle = \overline{\langle x, y\rangle}$$

   Importantly, this implies that $, x\rangle$ is a real number
2. The inner product is linear in its first argument. For all complex number $a, b \in \mathbb{C}$

$$\langle ax_1 + bx_2, y\rangle = a\langle x_1, y\rangle + b\langle x_2, y\rangle$$

3. The inner product of an element with itself is positive definite

$$\begin{aligned} \langle x, x\rangle &> 0 \qquad \text{if } x \neq 0 \\ \langle x, x\rangle &= 0 \qquad \text{if } x = 0 \end{aligned}$$

3.1.1 Principle 1: State vectors

The state of a physical system can be described (isolated from the rest of the universe) as a vector in a finite dimensional Hilbert space $\mathcal{H}$, we denote it as

$$|\psi\rangle \in \mathcal{H}$$

This vector $|\psi\rangle$ must be normalized with respect to the inner product of that Hilbert space, meaning $\langle\psi|\psi\rangle = 1$.

Remark 2. The principle doesn't say how to choose $|\psi\rangle$ neither the space itself $\mathcal{H}$. This decision comes from the underlying physics of the system.

Remark 3. In a non-isolated system we will have to generalize this principle. This will be done in later part of the course when introducing the notion of **density** matrix.

Hence let us start with the most basic example in quantum physics, the qbit! The Hilbert space describing a qbit is given as $\mathcal{H} = \mathbb{C}^2$ which can be denoted as

$$\begin{aligned}\mathcal{H} = \mathbb{C}^2 &= \left\{ \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \alpha, \beta \in \mathbb{C} \right\} \\ &= \{\alpha|0\rangle + \beta|1\rangle, \alpha \in \mathbb{C}, \beta \in \mathbb{C}\}\end{aligned}$$

Furthermore for this state to be correctly the state of a qbit it also must satisfy normalization, meaning its norm is 1. Recall not the definition of the inner product of a Hilbert space did we even define our inner product here? Recall that we need to validate the three properties above, a valid choice here is the following inner product:

$$\langle x, y\rangle = \sum_{i=1}^{n} x_i^* y_i$$

You can check that this inner product here fulfils all properties. Going back on our requirement ($\|x\| = 1$) we will therefore need

$$\alpha^*\alpha + \beta^*\beta = 1$$

With that requirement, we have the Hilbert space of a qbit.

Also in terms of notation, we will often use the letter ψ, φ and φ in order to represent quantum states. We will also use the term state to refer to quantum state as all work here is primarily done in the quantum world. Hence we define our qbit as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

For more general states, ones of dimension d (i.e. $\mathcal{H} = \mathbb{C}^d$) our state is constructed as

$$|\psi\rangle = \sum_{i=1}^{d} \alpha_i|i\rangle$$

Remark 4. Note that state doesn't have to be necessarily finite. A good example of this are quantum harmonic oscillators (so PHYS-344) in which our state is defined as

$$|\Psi\rangle = \sum_{n=0}^{\infty} \psi_n|n\rangle$$

3.1.2 Principle 2: Unitary evolution on time evolution

The state of an isolated system evolves (as a function of time) unitarily. This means that of $|\psi_0\rangle$ is a state at time $= 0$, the state $|\psi_t\rangle$ is given by

$$U_t|\psi_0\rangle = |\psi_t\rangle$$

Where $U_t^\dagger U_t = U_t U_t^\dagger = I$ i.e. U_t is unitary (usually time dependent) matrix

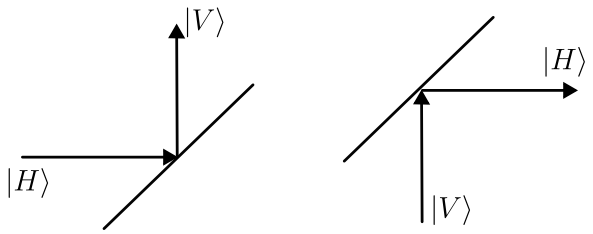
Remark 5. One can show that unitary matrix represent rotation of the space by a specified angles.

Remark 6. For people of have done or who are doing MATH-310 (Algebra), one can see that the time evolution matrices form a group in the sense that

$$U_{t_2}U_{t_1}U_{t_1+t_2} \qquad , U_{t=0} = I$$

Example: perfectly reflecting mirror

Let us take the example of a reflecting mirror :



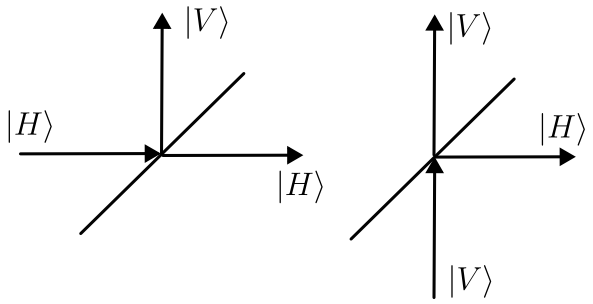
The physical process which transforms the incident ray into the reflected ray is described by the unitary matrix:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = |H\rangle\langle V| + |V\rangle\langle H|$$

For $|H\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|V\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Let us take a step back on what is actually going on here. What a perfectly reflecting mirror with a tilted angle here reflect a ray that is normal ($\langle\psi_{t_1}|\psi_{t_2}\rangle = 0$) for the state after and before the mirror, so here we would require that $U|H\rangle = |V\rangle$ and vis versa which if you apply X on any of those two states gives the intended results.

Semi-transparent mirror

Now let us define the same mirror but in this case we want '*half*' of the rays to reflect off the mirror and the other half to go through it. Given the initial state $|H\rangle$ the output that we are looking for is $\frac{|H\rangle+|V\rangle}{\sqrt{2}}$. On the other hand given a vector $|V\rangle$ we want the final state $\frac{|H\rangle-|V\rangle}{\sqrt{2}}$



The unitary matrix representing this operation is called the Hadamard matrix (gate) defined as

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Remark 7. One can find it using only the two conditions asked above. We require that $H|H\rangle = \frac{|H\rangle+|V\rangle}{\sqrt{2}}$ which implies that

$$H|H\rangle = \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle)\langle H|$$

We also require tat $H|V\rangle = \frac{|H\rangle-|V\rangle}{\sqrt{2}}$ giving

$$H|V\rangle = \frac{1}{\sqrt{2}(|H\rangle - |V\rangle)\langle|V\rangle|}$$

hence our matrix H is given by

$$\begin{aligned} H &= \frac{1}{\sqrt{2}}((|H\rangle + |V\rangle)\langle H| + (|H\rangle - |V\rangle)\langle V|) \\ &= \frac{1}{\sqrt{2}}(|H\rangle\langle H| + |V\rangle\langle H| + |H\rangle\langle V| - |V\rangle\langle V|) \\ &= \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \end{aligned}$$

3.1.3 Principle 3: Observable quantities

An observable (or measurable) quantity (such as energy, position, momentum, magnetic moment, polarization) is given by an Hermitian matrix A (i.e. $A^\dagger = A$)

Example

Let us take for instance the example of a simple qbit in the Hilbert space $\mathbb{C}^2$. It can be easily shown that any hermitian matrix are of the form

$$\begin{aligned} A &= \begin{pmatrix} \alpha & \beta \\ \overline{\beta} & \gamma \end{pmatrix} \\ &= \alpha|0\rangle\langle 0| + \beta|0\rangle\langle 1| + \overline{\beta}|1\rangle\langle 0| + \gamma|1\rangle\langle 1| \end{aligned}$$

Such 2×2 matrices can be represented as linear combination of so-called Pauli matrices

$$\begin{aligned} I &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ \sigma_x = X &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ \sigma_y = Y &= \begin{pmatrix} 0 & -i \\ 1 & 0 \end{pmatrix} \\ \sigma_z = Z &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \end{aligned}$$

Remark 8. Pauli matrices have a lot of useful properties and are used in everywhere in quantum physics (wiki), there are very useful as they represent projection of a qbit on the axis x, y, z when the qbit is represented on the Bloch sphere. Meaning that given the measurement of the qbit using this basis, one can deduce the projection of the qbit on those axis (on the Bloch sphere).

Exercises

One can show that any 2×2 matrices can be represented as a linear combination of those matrices, that is, for any A s.t. $A = A^\dagger$, A can be written as

$$A = a_0 I + a_x\sigma_x + a_y\sigma_y + a_z\sigma_z$$

With $a_0, a_x, a_y, a_z \in \mathbb{R}$.

3.1.4 Spectral theorem for Hermitian matrices

Let A be a $d \times d$ Hermitian matrix. Let $|v_1\rangle, \dots, |v_d\rangle$ and $\lambda_1, \dots, \lambda_d$ the eigenvectors and eigenvalues, i.e.

$$A|v_i\rangle = \lambda_i |v_i\rangle, \forall i = 1, \dots d$$

The eigenvectors can always be chosen to form an orthonormal basis of $\mathbb{C}^d$ and the eigenvalues $\lambda_1, \dots, \lambda_d$ are real ($\in \mathbb{R}$). Moreover, on can use the **spectral decomposition** giving

$$A = \sum_{i=1}^{d} \lambda_i \langle v_i || v_i \rangle$$

Remark 9. The last formula just says that in the basis $|v_1\rangle, \dots, |v_d\rangle$ of eigenvectors, A is diagonal with each λ_i at the A_{ii} positions.

Remark also that this definition is a lot useful here as it can be used to define functions applied to matrices. Let us for instance wanting to exponentiate our matrix A, meaning that we have a function f (the exponential) and seek to find $f(A)$. One can define (with functional analysis)

$$\exp(A) = \sum_{i=1}^{d} \exp(\lambda_i) |v_i\rangle\langle v_i|$$

This exponentiation here is pretty useful as one can therefore solve differential equation involving matrices (e.g. Schrödinger's equation).

3.1.5 Principle 4: Measurement postulate

let $|\psi\rangle$ be a state of a system. Let A be an observable that we measure.
The state of the measurement is a random outcome where the state becomes

$$|v_i\rangle \qquad \forall i = 1, \dots d$$

The '*value of A*' is λ_i (associated to $|v_i\rangle$)

And the probability law is

$$P(i) = |\langle v_i | \psi \rangle|^2$$

Given by the squared bracket between $|\psi\rangle$ and $|v_i\rangle$.

Second form of the measurement postulate often used in information theory

A measurement process is performed thanks to a 'measurement apparatus'. The measurement apparatus is modelled by an orthonormal basis of the Hilbert space $\{|v_1\rangle, |v_2\rangle, \dots, |v_d\rangle\}$ and the measurement outcome is a random state $|v_i\rangle$ with

$$P(i) = |\langle v_i | \psi \rangle|^2$$

Remark

The second form of the measurement postulate is used when we do not need to specify the observable being measured. Only the '*measurement apparatus*' is specified i.e. an orthonormal basis.

Property The probabilities in the measurement postulate sum to one, here is the proof

$$\begin{aligned}\sum_{i=1}^{d} P(i) &= \sum_{i=1}^{d} |\langle v_i|\psi\rangle|^2 \\ &= \sum_{i=1}^{d} \overline{\langle v_i|\psi\rangle}\langle v_i|\psi\rangle \\ &= \sum_{i=1}^{d} \langle\psi|v_i\rangle\langle v_i|\psi\rangle \\ &= \langle\psi|\left(\sum_{i=1}^{d} |v_i\rangle\langle v_i|\right)|\psi\rangle \\ &= \langle\psi|\psi\rangle = 1\end{aligned}$$

Here we have used the fact that $\sum_{i=1}^{d} |v_i\rangle\langle v_i| = I$.

3.1.6 Average value of an observable

Okay we have seen that we have now probability distribution over the eigenvectors of our observable, the question that this is raising is can we find a form of expected value? Well the answer is yes! and we can derive it ourself! For that let us recall the definition of the expected value in the classical case. Given a random variable X we define its expected value as (kind of)

$$\mathbb{E}[X] = \sum_{i=1}^{n} X_i P(X_i)$$

Here the probability are the defined $\overline{\langle v_i|\psi\rangle}\langle v_i|\psi\rangle$ and the result of the random variable are the λ_i hence our expected value would be defined as

$$\begin{aligned}\text{Av}(A) &= \sum_{i=1}^{d} \lambda_i \langle\psi|v_i\rangle\langle v_i|\psi\rangle \\ &= \langle\psi|\left(\sum_{i=1}^{d} \lambda_i |v_i\rangle\langle v_i|\right)|\psi\rangle \\ &= \langle\psi|A|\psi\rangle\end{aligned}$$

Which is a real value since $\lambda_i \in \mathbb{R}$ for any Hermitian matrix.

3.1.7 Heisenberg uncertainty principle

Let us define the mean square error (or the square root of the variance) of measurements for two observables A and B:

$$\Delta A = \sqrt{\langle\psi|A^2|\psi\rangle - \langle\psi|A|\psi\rangle^2}$$
$$\Delta B = \sqrt{\langle\psi|B^2|\psi\rangle - \langle\psi|B|\psi\rangle^2}$$

We have

$$(\Delta A)(\Delta B) \geq \frac{1}{2}|\langle\psi|[A,B]|\psi\rangle|$$

Where $[A,B] \equiv AB - BA$ (the commutator)

What this property says that the foundations of A and B cannot simultaneously vanish if they do not commute (i.e. $[A,B] \neq 0$).

3.1.8 Principle 5: Composition of systems

Let $\mathcal{H}_A$ be the Hilbert space of a system A and $\mathcal{H}_B$ be the Hilbert space of a system B. Then $\mathcal{H}_A \otimes \mathcal{H}_B$ is the Hilbert space of the composed system $A \bigcup B$. The state of the composed system is similarly a vector in $\mathcal{H}_A \otimes \mathcal{H}_B$.

Example Let us have the following spaces

$$\mathcal{H}_A = \mathbb{C}^2$$
$$\mathcal{H}_B = \mathbb{C}^2$$

With for instance $\mathcal{H}_A$ being the first qbit (e.g. first photon polarization) and $\mathcal{H}_B$ be the second qbit (e.g. second photon polarization). In that case states are of dimension 4 and live inside $\mathbb{C}^2 \otimes \mathbb{C}^2$.

Mathematical interlude, Tensor product

Before going down into composition of system, we need first to have notation for our new states, we need product states and entangled states.

Definition 9. (product states) Let $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$. It is said to be a **product state** if there exists $|\varphi_A\rangle \in \mathcal{H}_A, |\varphi_B\rangle \in \mathcal{H}_B$ such that

$$|\psi\rangle = |\varphi_A\rangle \otimes |\varphi_B\rangle$$

Definition 10. (Entangled states) A state $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ is said to be **entangled** if it is not a product state.

As we will see this classification of states plays a very important role in quantum information processing (e.g. quantum teleportation)

Examples

$$|0\rangle \otimes |0\rangle, \quad |0\rangle \otimes |1\rangle, |1\rangle \otimes |0\rangle, \ldots$$

Are product states, one can also construct linear combination of such

$$\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) \otimes \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) = \frac{1}{2}(|0\rangle \otimes |0\rangle + |0\rangle \otimes |1\rangle + |1\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$$

Is a product state.

$$\frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$$

On the other side is entangled (also called Bell state).

3.2 Geometrical representation of qbits and the Bloch sphere

Quantum state vectors belong to Hilbert space which consist of vectors with complex components. There are **note** vectors in usual Euclidean space and thus difficult to represent geometrically.
For the particular of **one qbit** where is a convenient geomtrical representation which we will often use an given good intuition (for two qbits the situation is already more complicated and for multiple qbits (large) this constitutes an open problem).
Hence, let us work through it for one qbit, recall that its space is $\mathcal{H} = \mathbb{C}^2$

$$\mathcal{H} = \{\alpha|0\rangle + \beta|1\rangle; \alpha, \beta \in \mathbb{C}\}$$

Where $|\alpha|^2 + |\beta|^2 = 1$. This implies that we have 4 real parameters with one constraints which implies that we only need 3 parameters to fully describe the space. In fact one should still remove one more parameter. The reason being that is the following :

$$|\psi\rangle \quad \text{and} \quad e^{i\gamma}|\psi\rangle \qquad , \gamma \in \mathbb{R}$$

are equivalent physically

Indeed γ can't move the observed! To get an intuition about this recall that the probabilities in the measurement postulate satisfy the following

$$|\langle v_i|\psi\rangle|^2$$

Now imagine that we where to shift the state by $e^{i\gamma}$ we would have the following probabilities

$$\begin{aligned}\left|\langle v_i|e^{i\gamma}|\psi\rangle\right| &= \left|e^{i\gamma}\right|^2|\langle v_i|\psi\rangle|^2 \\ &= |\langle v_i|\psi\rangle|^2\end{aligned}$$

In reality this implies that we should state principle 1 in a more fundamental form stating that

state vector of an isolated system are '*rays*' in the Hilbert space e.g. $e^{i\gamma}|\psi\rangle$, $\gamma \in \mathbb{R}$.

Remark 10. In $e^{i\gamma}|\psi\rangle$ γ is called a **global phase**. It **cannot be observed** but if you have $|0\rangle + e^{i\gamma}|1\rangle$ for example this is a **local phase**. And a local phase **can be observed** (e.g. in the interference experiments).

Then for any qbit represented by the state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, we can multiply by a global phase we arrive to make the component α real. What this gives is a grand total of 2 parameters eventually.

The canonical and useful parametrisation

So we have a vector of *norm* one and two free parameters. A good idea here would be to represent our state on a sphere based on those two parameters, let us define

$$\alpha = \cos\frac{\theta}{2}; \qquad \beta = \left(\sin\frac{\theta}{2}\right)e^{i\varphi}$$

Giving $\theta \in [0, \pi]$ and $\varphi \in [0, 2\pi]$.

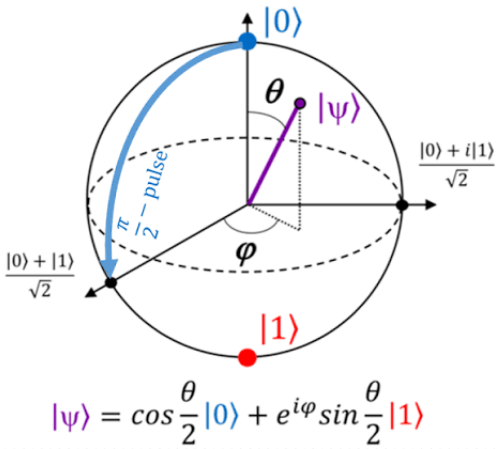


Figure 3.1: Representation of a qbit $|\psi\rangle$ on the Bloch sphere

Remark 11. Let us state here useful notation for spin interpretation which will be use later, we define the '*up state*' and '*down state*' as

$$|0\rangle = |\uparrow\rangle$$
$$|1\rangle = |\downarrow\rangle$$

Property Vectors that are orthogonal in $\mathbb{C}^2$ appear opposite on the Bloch sphere

For example let us take $\langle 1|0\rangle = 0$ and the same goes for $\langle +|-\rangle$.

And as said before, unitary evolution present the norm of vectors. This implies that on the Bloch sphere **unitary evolution will appear as rotations of vectors on this sphere** (more on this later).

3.3 Application of the interference experiments

The goal of this section if to illustrate the application of the principle to a few interference experiments. This will allows to better understand them on one hand, as well as have concrete examples of qbits, before we go on to communication protocols in the next few sections.

3.3.1 Example of qbits

Let us take the example of path states inthe Mach-Zehender experiments. Recall the simple MZ interferometer of the first lecture : Where the rectangle here are semi-transparent mirror and lines are

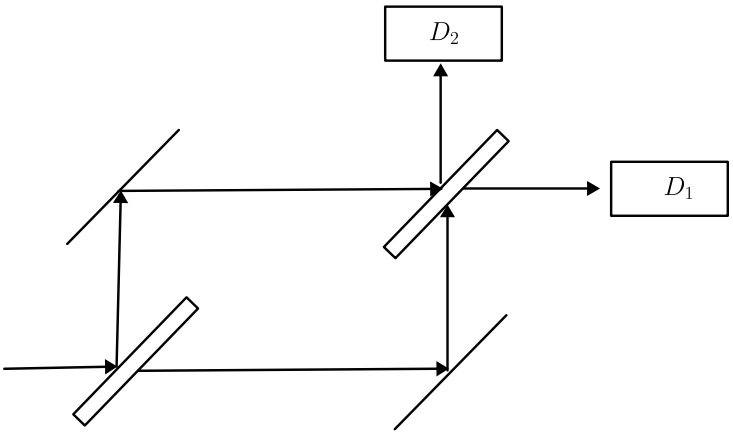


Figure 3.2: MZ interferometer experiment

reflecting mirror. A '*toy*' model for the Hilbert space is $\mathcal{H} = \mathbb{C}^2 = \{\alpha|H\rangle + \beta|V\rangle\}$ where

$$|H\rangle = \begin{pmatrix}1\\0\end{pmatrix} = \text{ horizontal path state of photon}$$
$$|V\rangle = \begin{pmatrix}0\\1\end{pmatrix} = \text{ vertical path state of photon}$$

With $\alpha, \beta \in \mathbb{C}$ such that $|\alpha|^2 + |\beta|^2 = 1$.

Polarization of photons Photons carry with them a **polarization** vector which is analogous to the polarization of the electromagnetic vector. We will see that this is described by vectors of $\mathbb{C}^2 = \mathcal{H}$. Thus the polarization of photons is a quantum bit. The description of terms of qbit is '*exact*' this time (i.e. this is not a toy model approximation).

Spin $\frac{1}{2}$ Many particles (e.g. electrons, photons, some nuclei) carry a small magnetic dipole whose state vector is described by a vector in $\mathcal{H} = \mathbb{C}^2$.
This is analogue (but different) to polarization of photons. We will come back to spin later in the class when we sill study its interaction with magnetic fields. Again here we have an '*exact qbit*'.

Two level systems Just like atoms have energy level corresponding to different orbital states, many systems are described by some net of '*energy level*'.

Often one can isolate how important levels from the set of spectrum and model the system to a good approximation by e qbit with Hilbert space $\mathbb{C}^2$.

Hence after this parenthesis, let us go back to our experiment and try to analyse it.

3.3.2 Analysis of MZ interferometer 1

Let us recall the experimental setup, first the Hilbert space: $\mathcal{H} = \mathbb{C}^2 = \{\alpha|h\rangle + \beta|v\rangle, \alpha, \beta \in \mathbb{C}\}$. Where $|h\rangle = \begin{pmatrix}1\\0\end{pmatrix}, |v\rangle = \begin{pmatrix}0\\1\end{pmatrix}$. Here the state represent the '*path state*' of the given photon. Here the path state must also satisfy the normalization of the vector

$$|\alpha|^2 + |\beta|^2 = 1$$

The input state here is simply $|h\rangle$. The goal here is to see how this input state evolve in time. To do so let us recall the second principle which state that the evolution is given by a unitary matrices (when not measured). Hence the first semi-transparent mirror is represented by the **Hadamard** matrix:

$$\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\1 & -1\end{pmatrix} = \frac{1}{\sqrt{2}}(|h\rangle\langle h| + |h\rangle\langle v| + |v\rangle\langle h| - |v\rangle\langle v|)$$

And as we can see we effectively have $HH^\dagger = H^\dagger H = I$.

The output of the semi-transparent mirror is given by

$$H|h\rangle = \frac{1}{\sqrt{2}}(|h\rangle + |v\rangle) = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$$

Perfect mirror

Let us continue with the second principle and see what the perfect mirrors do to our current quantum state. To see what they do let us first define them

$$X = \begin{pmatrix}0 & 1\\1 & 0\end{pmatrix} = |h\rangle\langle v| + |v\rangle\langle h|$$

As we can see, this unitary matrix behave as a logical negation meaning taking $X|h\rangle$ gives $|v\rangle$ and vis versa.

Output of the reflection

Now let us see what is the output of the two mirrors together

$$XH|h\rangle = \frac{1}{\sqrt{2}}(X|h\rangle + X|v\rangle) = \frac{1}{\sqrt{2}}(|v\rangle + |h\rangle)$$

Finally, let us see what is the final state after the final semi-transparent mirror. As the first mirror, this one is also represented using the same unitary matrix H. To see the final state, we as before are required to apply H to the current state which gives the following

$$HXH|h\rangle = \frac{1}{\sqrt{2}}(H|v\rangle + H|h\rangle) = \frac{1}{\sqrt{2}}\left(\frac{|h\rangle - |v\rangle}{\sqrt{2}}\frac{|h\rangle + |v\rangle}{\sqrt{2}}\right) = |h\rangle$$

Remark The wave interpretation here would explain this as making the D_2 output being a destructive wave addition wheras the output into D_1 is constructive.

Principle 4

Recall the principle: *Measurement is done in a basis*, we will use $\{|h\rangle, |v\rangle\}$. Using this basis, one can use Born rule to compute the probability of each one being selected as

$$P(|h\rangle \text{ being selected}) = P(D_1, \text{clicks}) = |\langle h|h\rangle|^2 = 1$$
$$P(|v\rangle \text{ being selected}) = P(D_2, \text{clicks}) = |\langle v|h\rangle|^2 = 0$$

Remark 12. Let us first recall principle 3 which states that an **Observable** or quantity that I measure can be represented using a hermitian operator where each eigenvalues are an output and its correspondent eigenvector be the vector to compute the probability of the given output.

In our case the observable which can be called C is defined the following way

$$C = (+1)|h\rangle\langle h| + (-1)|v\rangle\langle v| = \begin{pmatrix} +1 & 0 \\ 0 & -1 \end{pmatrix}$$

Remark 13. The whole experiment has a '*circuit interpretation*'. Quantum circuit constitute a representation of quantum algorithms (see in the second semester CS-308)

3.3.3 MZ interferometer 2

Let us look at the following variation of the preceding interferometer :

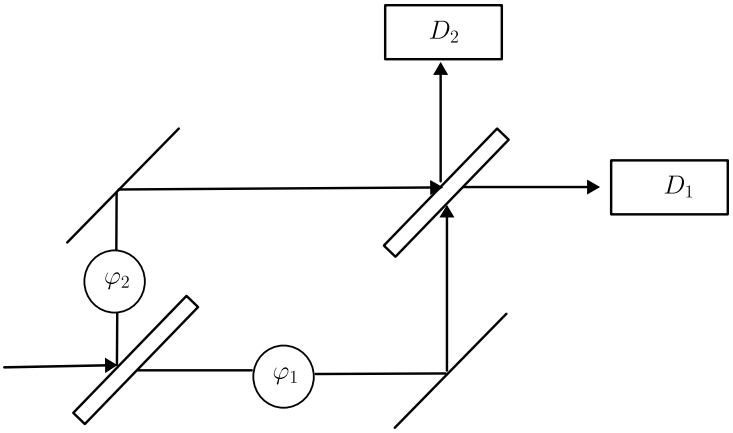


Figure 3.3: MZ interferometer 2

- φ_1 is a '*dephaser*' which changes the '*phase*' of a photon by $|H\rangle \to e^{i\varphi_1}|H\rangle$
- β_2 is a '*dephaser*' which changed the '*phase*' of a photon by $|V\rangle \to e^{i\varphi_2}|V\rangle$

Their corresponding unitary operation are

$$P_1 = \begin{pmatrix} e^{i\varphi_1} & 0 \\ 0 & 1 \end{pmatrix}$$
$$P_2 = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\varphi_2} \end{pmatrix}$$

Now for the perfectly reflecting mirrors and semi-transparent one can choose the following operator in order to model them :

$$U_{st} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$$
$$U_r = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

As we can check, both P_i matrices here share the same eigenbasis which implies that they are commutative which mean that the evolution in time can be computed the following way

$$U_{st}U_rP_1P_2U_{st}|h\rangle = -\frac{e^{i\varphi_2} + e^{i\varphi_2}}{2}|h\rangle + i\left(\frac{e^{i\varphi_1} - e^{i\varphi_2}}{2}\right)|v\rangle$$

Hence the probabilities are computed as

$$\begin{aligned} P(D_1 \text{ clicks}) = |\langle H||\psi_{\text{final}}\rangle|^2 &= \left|\frac{e^{i\varphi_1}+e^{i\varphi_2}}{2}\right|^2 \\ &= \frac{\left|1+e^{i(\varphi_2-\varphi_2)}\right|^2}{4} \\ &= \frac{1}{4}\Big[(1+\cos(\varphi_2-\varphi_1))^2+\sin^2(\varphi_1-\varphi_2)\Big] \\ &= \frac{1}{2}(1+\cos(\varphi_2-\varphi_1)) \\ &= \left[\cos\left(\frac{\varphi_1-\varphi_2}{2}\right)\right]^2 \\ &= \cos^2\left(\frac{\varphi_1-\varphi_2}{2}\right) \end{aligned}$$

Similarly we compute

$$\begin{aligned} P(D_2 \text{ clicks}) = |\langle V||\psi_{\text{final}}\rangle|^2 &= \frac{1}{4}\left|e^{i\varphi_1}-e^{i\varphi_2}\right|^2 \\ &= \frac{1}{4}\left|1-e^{i(\varphi_2-\varphi_1)}\right|^2 \\ &= \frac{1}{4}\Big[(1-\cos(\varphi_2-\varphi_1))^2+(\sin(\varphi_2-\varphi_1))^2\Big] \\ &= \frac{1}{2}(1-\cos(\varphi_2-\varphi_1)) \\ &= \left(\sin\left(\frac{\varphi_1-\varphi_2}{2}\right)\right)^2 \end{aligned}$$

In this case, we can also define the following observable matrix

$$C' = (+1)|+\rangle\langle+| + (-1)|-\rangle\langle-|$$

3.3.4 From electromagnetic to quantum mechanics

Photon polarization Electromagnetic waves carry a degree of freedom called '*polarization*'

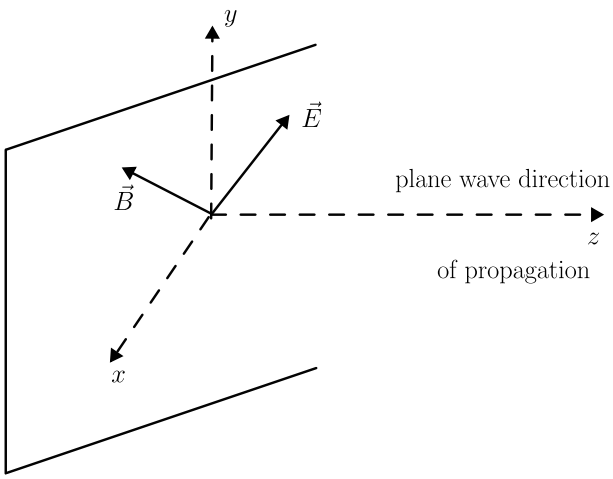


Figure 3.4: Electromagnetic Wave

Where the (x, y) place is orthogonal to z. We have that the $\left(\vec{E}, \vec{B}\right)$ fields oscillate in the plane perpendicular to the direction of propagation. This can be modelled as

follow

$$\vec{E}(z,t) = \mathrm{Re}\Big\{\vec{E}_0 e^{i(kz-\omega t)}\Big\}$$

Where $k = \frac{2\pi}{\lambda}, \omega = 2\pi\nu$, where $\lambda =$ wavelength, $\nu =$ frequency. With finally

$$\vec{E}_0 = \underbrace{E_0}_{\in\mathbb{R}} \begin{pmatrix} \cos\theta \\ \sin\theta e^{i\varphi} \\ 0 \end{pmatrix}$$

Linear polarization

In the case $\varphi = 0$ the $\vec{E}-$field oscillates along θ direction in the (x,y) plane, I don't have the strength to do all diagram of the course here so for this part I will invite you to check out the lecture notes, Chapter 2.1-2.4 of notes

The way we have defined our qbit only took into account two directions. However why should we limit ourself to this? Polarization degree of freedom of a photons, spin of a particles can also be perfectly represented as qbits.

Electromagnetic field

First let us recall what is an electromagnetic field is, using Maxwell equations, the field can be defined using a magentic and electric field $\vec{B}$ and $\vec{E}$ respectively which vibrate into a plane normal to a given $\vec{z}$ direction. Using Maxwell equation, one can deduce $\vec{B}$ using only $\vec{E}$. Hence the field can be defined only using $\vec{E}$ which cen be defined the following way

$$\vec{E} = \mathrm{Re}\Big\{\vec{E}_0 e^{i(kz-\omega t)}\Big\}$$

Where the $\vec{E}_0$ set the direction of the vibration of the field and the exponent defines the polarization. Which when taking this into a vector notation, we obtain

$$E_0 \begin{pmatrix} \cos\theta \\ \sin\theta e^{i\varphi} \\ 0 \end{pmatrix}$$

Photons also retain a sort of '*pole state*' defined as $\begin{pmatrix} \cos\theta \\ \sin\theta e^{i\varphi} \end{pmatrix} \in \mathbb{C}^2 = \mathcal{H}$. As we can see here, we would like to define our state using the x-y plane inside a plane. However this $e^{i\varphi}$ gives us a problem here, what this factor implies physically is that the photon also disturb the $z-$plane.

To translate this into Dirac notation, we define it the following way

$$\cos\theta|0\rangle + \sin\theta e^{i\varphi}|1\rangle$$

Example

If for instance we define the case where $\varphi = \frac{\pi}{2}$ and $\theta = \frac{\pi}{4}$ we get the following state

$$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ \pm i \end{pmatrix}$$

Which as we can see implies that the polarization given by the $e^{i\varphi}$ term can either rotate in both direction. Here as $\varphi \neq 0$, we have that the blabla is rotating around the z axis. The φ given here represent the polarization degree of freedom of our photon.

Chapter 4

Quantum Cryptography

4.1 Quantum Key Distribution

This section is about quantum key distribution (QKD) for the **generation of common one-time pad shared by two distant parties Alice and Bob**. The goal is to use quantum information in order to reliably share bits between two or ore distant parties. The goal is to do this while insuring privacy of the bit being transmitted, to do so, we use qubit which have the property of non-cloning. If you have done last week exercise, you will know that given any non-orthogonal state, there is no unitary operation for which one can copy the state. Formally this can be written as there is no unitary operator $U \in \mathcal{H}$ which for all state $|\psi\rangle$ gives

$$U|\psi\rangle \otimes |\text{blank}\rangle = |\psi\rangle \otimes |\psi\rangle$$

An even more general proposition can be done which state that

In any attempt to distinguish between two non-orthogonal quantum states, information gain is only possible at the expense of introducing disturbance to the signal.

(This is taken from Nielsen and Chuang, the proof is also there (Sec. 12.6.3)).

What this implies is that one can use that fact in order to protect the states: One can create redundant qubits which we will serve as a test in which Alice and Bob will check for any disturbance of the states if they found disturbance (outside of errors), then Alice and Bob know that there has been an eavesdropper. Hence it is important to have non-orthogonality in the state being transmitted.

Remark 14. Let us first recall here what one-time-pad is. This is a sequence of secret bits $z_1, z_2, \ldots, z_m$, $z_i \in \{0, 1\}$ shared by Alice and Bob (which are supposed to be at two distant locations). It is used as follows:

- If Alice has a message $m_1 m_2 \ldots m_M$ to communicate, she encodes it as

  $$(z_1 \oplus m_1)(z_2 \oplus m_2)\ldots(z_M \oplus m_M)$$

  and sends the sequence to Bob. Then Bob can decode it as

  $$(z_i \oplus m_i) \oplus z_i = m_1$$

  This scheme of encrypting is theoretically a secure scheme if the one-time pad is i.i.d. (independent identically distributed) uniformly random and is used only once.
- The trouble is how to distribute it to Alice and Bob and make sure it is not intercepted

The point of the QKD is that we do not distribute the one-time pad but generate it directly in the labs of Alice and Bob.

4.1.1 BB84 protocol for QKD

Here let us review the main phases of the protocol

1. Encoding phase of Alice
2. Decoding phase of Bob
3. Public communication phase of Alice and Bob
4. Generation of common secret bit sequence and the '*security check*'

Finally, we will review some possible attacks from eavesdroppers and argue whether the protocol is secure.

1: Encoding in Alice's lab

- Alice generates two uniformly random i.i.d. sequences of classical bits

$$x_1 x_2 \dots x_N \in \{0,1\}^N$$
$$e_1 e_2 \dots e_N \in \{0,1\}^N \quad \text{secret for the moment}$$

- For $e_i = 0$ Alice prepares a qbit $|x_i\rangle \in \{|0\rangle, |1\rangle\}$ (in the computational basis)
- For $e_i = 1$, Alice prepares a qbit

$$H|x_i\rangle \in \left\{ \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right\}$$

In the Hadamard basis $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Remark 15. Physically you can think about polarized states of photons for the qbits.

To summarize, at each instant $i = 1, \dots, N$ Alice has prepared a qbit in the state

$$H^{e_i}|x_i\rangle$$

Remark The power here $e_i \in \{0,1\}$ just say if $e_i = 1$ then apply H, which just a notation trick.

Hence we have that the qbits we are sending are inside the set

$$H^{e_i}|x_i\rangle \in \left\{ |0\rangle, |1\rangle, \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right\}$$

A then sends this qbit to Bob through a quantum channel (optic fibers, free space etc.)

Decoding in Bob's lab

Bob receives the qbit (photon polarization) for each $i = 1, \dots, N$. He has no idea of the state. He does a measurement at each instant.

- Bob generates an i.i.d. uniformly random sequence of bits

$$d_1 d_2 \dots d_N \in \{0,1\}^N$$

- For $d_i = 0$ he choose the measurement basis $\{|0\rangle, |1\rangle\}$ to do his measurement
- For $d_i = 1$ he choose the measurement basis $\left\{ \frac{|0\rangle+|1\rangle}{\sqrt{2}}, \frac{|0\rangle-|1\rangle}{\sqrt{2}} \right\}$

In summary, his output state is $H^{d_i}|y_i\rangle$

3: Public communication phase

Now Alice and Bob share publically their encoding and decoding sequences $e_1, \dots e_N$ and $d_1, \dots d_N$. Anybody may know these sequences. However $x_1, \dots, x_N$ and $y_1, \dots, y_N$ are **not** revealed

4: Generation of one-time pad

- If $e_i = d_i$, Alice and Bob keep the bits x_i and y_i. It turns out that $x_i = y_i$
- If $e_1 \neq d_i$, then Alice and Bob discard the bits x_i and y_i. It turns out thats those may be different or equal

Security check

out of $i = 1 \dots N$ such that $e_i = d_i$ Alice and Bob select $\frac{\varepsilon N}{2}$ such instances ($0 < \varepsilon \ll 1$) and sacrifice the (x_i, y_i) by echanging then on the public channel. Thus they check if indeed $x_i = y_i$. If this let passes they conclude that the protocol has worked well and nobody has been eavesdropping.
Essentially the security check lets if

$$\frac{1}{\left(\frac{\varepsilon N}{2}\right)}\left\{ i \in \left[\frac{\varepsilon N}{2}\right] \subset \{1, \dots, N\} \quad \text{s.t. } e_i = d_i \mid x_i = y_i \right\} \approx 1$$

Remark that here the noise of the channel should be specified and low enough so that we are able to distinguish noise from eavesdroppers.

Lemma

This whole protocol is justified by the following lemma:

$$\begin{cases} P(x_i = y_i | e_i = d_i) = 1 \\ P(x_i \neq y_i | e_i = d_i) = 0 \end{cases}$$

$$\begin{cases} P(x_i = y_i | e_i \neq d_i) = \frac{1}{2} \\ P(x_i \neq y_i | e_i \neq d_i) = \frac{1}{2} \end{cases}$$

Proof

Let us compute the first point of the first case $P(x_i = y_i | e_i = d_i)$. Here we can separate it into two cases

$$\begin{aligned} P(x_i = y_i | e_i = d_i) &= P(x_i = 0, y_i = 0 | e_i = d_i) + P(x_i = 1, y_1 = 1 | e_i = d_i) \\ &= P(y_i = 0 | x_i = 0, e_i = d_i) P(x_i = 0) + P(y_i = 1 | x_i = 1, e_i = d_i) P(x_i = 1) \end{aligned}$$

Then using Bayes law we have $P(A|B) = \frac{P(A \cap B)}{B} \implies P(A \cap B) = P(A|B)P(B)$. Now what is the state given to Bob when $x_i = 0$? It is $H^{e_i}|0\rangle$. If $e_i = 0$ then we have sent $|0\rangle$ which as $d_i = e_i = 0$ Bob will measure using the basis $\{|0\rangle, |1\rangle\}$ output $y_i = 0$ with probability 1. On the other side if $e_i = d_i = 1$ then Alice will have sent $|+\rangle$ and Bob will measure it using the basis $\{|+\rangle, |-\rangle\}$ which will also output $y_i = 0$ with probability 1. The same measurement can be done when $y_i = 1$ giving

$$\begin{aligned} P(x_i = y_i | e_i = d_i) &= P(x_i = 0) + P(x_i = 1) \\ &= \frac{1}{2} + \frac{1}{2} = 1 \end{aligned}$$

Hence $P(x_i \neq y_i | e_i = d_i) = 1 - P(x_i = y_i | e_i = d_i) = 0$.

Let us now look at the other case

$$\begin{aligned} P(x_i = y_i | e_i \neq d_i) &= P(x_i = 0, y_i = 0 | e_i \neq d_i) + P(x_i = 1, y_i = 1 | e_i \neq d_i) \\ &= P(y_i = 0 | e_i \neq d_1, x_i = 0) P(x_i = 0) + P(y_i = 1 | e_i \neq d_i, x_i = 1) P(x_i = 1) \end{aligned}$$

If $e_i \neq d_i$, we have that the state sent to Bob is on the other basis that the one it will be measured with. Meaning that if the state sent is $|0\rangle$ then it will be measured with the $\{|+\rangle, |-\rangle\}$ basis hence the probability of having $y_i = 0$ is given as

$$|\langle + || 0 \rangle|^2 = \frac{1}{2}$$

Now if the state sent is $|+\rangle$ which implies that $e_i = 1$ then $d_i = 0$ which therefore mean that we will measure using the basis $\{|0\rangle, |1\rangle\}$ hence giving the probability that $y_i = 0$ as

$$|\langle 0 || + \rangle|^2 = \frac{1}{2}$$

Which implies that $P(y_i = 0 | e_1 \neq d_i, x_i = 0) = \frac{1}{2}$ and the same goes for the $y_i = 1$ case giving

$$P(y_i = y_i | e_i \neq d_i) = \frac{1}{2}\frac{1}{2} + \frac{1}{2}\frac{1}{2} = \frac{1}{2}$$

4.1.2 Attack from an eavesdropper

A complete discussion of security of the scheme goes well beyond this course. One must imagine that an eavesdropper captures photon on the quantum channel, process them (with unitary evolutions and measurements), sends them back to Bob etc. The space of possible attack is huge and here we just scratch the surface of this subject by analyzing a very simple type of attack.

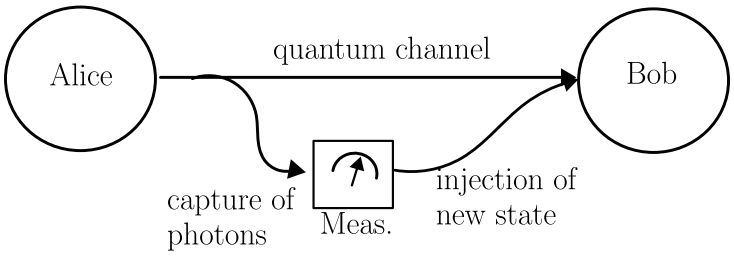


Figure 4.1: Setup of the eavesdropper attack

Measurement attack

Let us imagine the following setup

- Eve captures photons in state $H^{e_i}|x_i\rangle$
- Measures in computational or Hadamard basis according to a random sequence she generates

$$E_1 E_2 \dots E_N \in \{0,1\}^N$$

  Note that she has no idea of e_i, d_i before public communication phase.
- The measurement learn the photon in state

$$H^{E_i}\left|y_i^{\text{Eve}}\right\rangle \quad ; y_i^{\text{Eve}} \in \{0,1\}$$

- The state $H^{E_i}|y_i\rangle^{\text{Eve}}$ is forwarded to Bob.
- Bob measures as usual (he doesn't know what happens on the quantum channel) and he gets a state $H^{d_i}|y_i\rangle$ as before but this time with probability

$$\left|\langle y_i|H^{d_i}H^{E_i}\left|y_i^{\text{Eve}}\right\rangle\right|^2$$

We now have the following lemma

Lemma

$$\begin{cases} P_{\text{Eve}}(x_i = y_i | e_i = d_i) & = \frac{3}{4} \\ P_{\text{Eve}}(x_i \neq y_i | e_i = d_i) = \frac{1}{4} \end{cases}$$

Where P_{Eve} mean the probability calculated in presence of Eve as in the sense of the probability of the decoding being '*right*' For Alice and Bob when Eve is disturbing everything.

This lemma implies that the security protocols is not passed **after** the public communication phase since $\frac{1}{4}$ of potential one-time-pad is corrupted:

$$\frac{1}{\frac{\varepsilon N}{2}}\left\{ i \in \left[\frac{\varepsilon N}{2} \subset [1,\dots,N] \text{ s.t. } e_i = d_i | x_i = y_i\right]\right\} \approx \frac{3}{4} < 1$$

Proof

Now let us compute as before:

$$P_{\text{Eve}}(x_i = y_i | e_i = d_i) = P(x_i = y_i | e_i = d_i, E_i = e_i)P(E_i = e_i) + P(x_i = y_i | e_i = d_i, E_i \neq e_i)P(E_i \neq e$$

In the first case here we have '*correctly guessed*' the state sent by Alice hence we resend the same state back to Bob giving that the probability of a good state is 1. On the other hand we have incorrectly guessed the basis in which the state has been

sent to Bob which therefore means that the state sent to Bob is in the wrong basis This implies that that probability is equivalent to

$$= P(x_i = y_i | e_i = d_i)P(E_i = e_i) + P(x_i = y_i | e_i \neq d_i)P(E_i \neq e_i)$$

A more formal way than before to compute the first term here is using Born rule on the state sent to Bob and the eigenvector of the measurement

$$\begin{aligned} P(x_i = y_i | e_i = d_i, E_i = e_i) &= P\big(y_i^{\text{Eve}} = y_i | E_i = d_i\big) \\ &= \left| \langle y_i | \underbrace{H^{d_i} H^{E_i}}_{=I} | y_i^{\text{Eve}} \rangle \right|^2 \\ &= \left| \langle y_i | y_i^{\text{Eve}} \rangle \right|^2 = 1 \end{aligned}$$

On the right hand side given $E_i \neq e_i$ we have $y_i^{\text{Eve}} = x_i$ with probability $\frac{1}{2}$ Thus

$$P(x_i = y_i | e_i = d_i, E_i \neq e_i) = P()$$

The question that this is raising is what does Eve wins other than disturbing Bob's and Alice's key?

Recall that the attack can be stated as follow, Eve takes the qbits from Alice and does a random measurement on them, and send the resulting state to Bob, therefore after Alice and Bob have shared they bit sequences, she know exactly the state sent to Bob and also d_i and e_i hence when d_i matches E_i Eve have access with certainty the bit that Bob has decoded As the probability is computed as

$$\left| \langle y_i | \underbrace{H^{d_i} H^{E_i}}_{=I} | y_i^{\text{Eve}} \rangle \right|^2 = 1$$

If $e_i = d_i = E_i$ then Alice, Bob and Eve share the same bit with certainty. When $e_i = d_i \neq E_i$ then Eve knows that Bob has a resulting that is right with probability $\frac{1}{2}$. The reasoning can be done as follow:
As $e_i \neq E_i$ Eve measure with the wrong basis giving equal probability to both state sent to Bob. For instance if she receives $|0\rangle$ from Alice then with equal probability she will send $|+\rangle$ and $|-\rangle$ to Bob. As Bob as $e_i = d_i$ He will measure using the other basis that the one used by Eve hence having equally likely result for both outcome which implies that there is a probability $\frac{1}{2}$ of Bob having the wrong bits while thinking he has a matching one with Alice.
However this is already '*good*' but this is not where Eve gains the most. Recall what happens when $E_i = e_i$, in that case, Eve perfectly decodes the state and resend it back to Bob. If $d_i = e_i = E_i$ then all three share the bits with certainty which could be useful if there weren't 25% of errors.

Impossibility of '*copy photon*' and resend attack

One could say here, well Eve could just copy the qbit that she received, send one to Bob and measure on her. What this would require is that we have a machine which create a copy of a qbit based on another qbit. This machine can be represented as a unitary operator hence we would have

$$U(|\psi\rangle \otimes |\text{blank}\rangle) = |\psi\rangle \otimes |\psi\rangle$$

Which if you have done homework 3 would know that this is not possible for $|\psi\rangle$ that are non orthogonal. And in fact here recall that our states can take all of the following values

$$\left\{ |0\rangle, |1\rangle, \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right\}$$

Hence we are protected from copying by the no cloning theorem.

Chapter 5

Quantum Entanglement

5.1 Concept of Entanglement

Let us define our total Hilbert space as $\mathcal{H}_{\text{total}} = \mathcal{H}_a \otimes \mathcal{H}_b$ (e.g. two quantum bits together $\mathcal{H}_a = \mathbb{C}^2, \mathcal{H}_b = \mathbb{C}^2$). Let us also recall the definition of the product state

Definition 11. (Product state) Product state $|\varphi\rangle$ are such that

$$\exists |\varphi_a\rangle \in \mathcal{H}_a, |\varphi_b\rangle \in \mathcal{H}_b$$

such that $\mathcal{H}_{\text{total}} \in |\varphi\rangle = |\varphi_a\rangle \otimes |\varphi_b\rangle$

What we mean is that a state is a product state if our state $|\varphi\rangle$ can be written as a tensor product between to state from each Hilbert space $\mathcal{H}_a$ and $\mathcal{H}_b$. Recallig also the definition of Entangled state

Definition 12. (Entangled states) **Entangled states** are states which cannot be written as a tensor product of two state, state that are not product state.

Example

If we take for instance the following state inside $\mathbb{C}^2 \otimes \mathbb{C}^2$:

$$\begin{aligned} |\varphi\rangle = |+\rangle \otimes |-\rangle &= \frac{|0\rangle + |1`\rangle}{\sqrt{2}} \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}} \\ &= \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle) \end{aligned}$$

Here we have that $|\varphi\rangle$ is a product state by construction.

Let us for instance change a minus sign in our state giving

$$|\psi\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle + |11\rangle)$$

To prove that the state $|\psi\rangle$ is not a product state, let us use contradiction. Assume that our state is a product state, then we know that there must exists four coefficients such that

$$|\psi\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle)$$

Here one can just develop our four equalities (one for each $|ij\rangle$) giving us the following system of equations

$$\begin{cases} \alpha\gamma = \frac{1}{2} \\ \alpha\delta = -\frac{1}{2} \\ \beta\gamma = \frac{1}{2} \\ \beta\delta = \frac{1}{2} \end{cases}$$

If we were working with real number, this can be proven fairly easily as the product of the LHS would be positive whereas the RHS would me negative ($-\frac{1}{16}$). But as we are working with complex number one should show it differently, what we need to do here is to make the modulus pop out. To do so, one can just take the conjugate as we have that $\overline{\alpha\gamma} = \frac{1}{2}$ and that $\overline{\beta\delta} = \frac{1}{2}$. Which when now multiplying everything together gives

$$|\alpha|^2|\beta|^2|\gamma|^2|\delta|^2 = -\frac{1}{16}$$

Which is a contradiction.

Hence here we indeed find that some states cannot be written as product of other, so we have a sort of duality, the question that this arise is can we '*quantify*' how much are a state entangled? We will be able to answer this question later in the course, however we will first introduces some states that we will see later are the '*most*' entangled ones.

5.1.1 Bell states

There are four states that plays a special roles in quantum communication and information theory, in teleportation, dense coding but also in QKD and many more topics. Those state are the '*maximum entangled*'

What are those? The bell states form an orthogonal basis of the state of two qbits $\mathbb{C}^2 \otimes \mathbb{C}^2$. Those state are given as

$$|B_{00}\rangle = \frac{1}{\sqrt{2}}(|0\rangle_A \otimes |0\rangle_B + |1\rangle_A \otimes |1\rangle_B)$$
$$|B_{01}\rangle = \frac{1}{\sqrt{2}}(|0\rangle_A \otimes |1\rangle_B + |1\rangle_A \otimes |0\rangle_B)$$
$$|B_{10}\rangle = \frac{1}{\sqrt{2}}(|0\rangle_A \otimes |0\rangle_B - |1\rangle_A \otimes |1\rangle_B)$$
$$|B_{11}\rangle = \frac{1}{\sqrt{2}}(|0\rangle_A \otimes |1\rangle_B - |1\rangle_A \otimes |0\rangle_B)$$

Here, one can show here that those states are entangled.

Computational orthogonal basis Let us also define the computational orthogonal basis of our new space $\mathbb{C}^2 \otimes \mathbb{C}^2$ namely

$$|0\rangle \otimes |0\rangle = \begin{pmatrix}1\\0\\0\\0\end{pmatrix} \qquad |0\rangle \otimes |1\rangle = \begin{pmatrix}0\\1\\0\\0\end{pmatrix}$$
$$|1\rangle \otimes |0\rangle = \begin{pmatrix}0\\0\\1\\0\end{pmatrix} \qquad |1\rangle \otimes |1\rangle = \begin{pmatrix}0\\0\\0\\1\end{pmatrix}$$

Important property of $|B_{00}\rangle$ Let us define the state $|\theta\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle$ which is the polarization state of a photon by θ. Then we can also define the perpendicular state

$$|\theta_\perp\rangle = -\sin\theta|0\rangle + \cos\theta|1\rangle$$

Theorem 2.

$$|B_{00}\rangle = \frac{1}{\sqrt{2}}(|\theta\rangle_A \otimes |\theta\rangle_B + |\theta_\perp\rangle \otimes |\theta_\perp\rangle_B)$$

What this implies is that the entanglement here $|B_{00}\rangle$ is angle independent. In Alice's lab, the angle of our state locally is sort of '*random*' but is the **same** as the

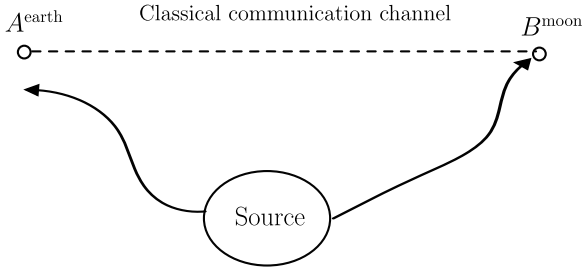
state inside Bob's lab. What we mean here by that which we will see later is that the entropy of our angle in both labs are maximized.

Remark The prove for all $\theta \in \mathbb{R}$ is left to the reader.

5.1.2 Teleportation

Assume that we have Alice A^{earth} on earth which has a photon in a state $|\varphi\rangle_1 = \alpha|0\rangle_1 + \beta|1\rangle_1$ where $\alpha, \beta \in \mathbb{C}$ are unknown. Let us assume that are another state $|B_{00}\rangle$ in our system. The total composed system is then given as $|\varphi\rangle_1 \otimes |B_{00}\rangle_{23}$. Assume now that we have a source creating those two state ($|\varphi\rangle_1$ and $|B_{00}\rangle$) connected to Alice and Bob. Also assume that we have a classical communication channel between Alice and Bob :

Protocol



Given our initial state $|\varphi\rangle_1 \otimes |B_{00}\rangle_{23}$ the goal of the protocol is to '*teleport*' the state $|\varphi\rangle$ from Alice to Bob, $|\varphi\rangle_1|\varphi\rangle_3$ where we have the entangled state $|B_{00}\rangle$ being split with one qbit to Alice $|\varphi\rangle_2$ and the other to Bob $|\varphi\rangle_3$ (note that we say separate in the sense that one qbit is at Alice's location and the other at Bob's location and not as two separate states). The protocol for teleportation is given as follow :

First step Alice does a local measurement of her two qbits with apparatus that's equivalent to a Bell basis.

Second step Alice observes her result, one among $|B_{00}\rangle, |B_{01}\rangle, |B_{10}\rangle, |B_{11}\rangle$ and sends two classical bits of information through the classical channel.

Third step Bob receives the classical message $ij \in \{00, 01, 10, 11\}$ and will do a local unitary operation on his qbit.

- If he receives $00 \to I_3$
- $01 \to X_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
- $10 \to Z_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
- $11 \to Z_3X_3 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

Then Bob receives exactly $|\varphi\rangle_3 = \alpha|0\rangle_3 + \beta|1\rangle_3$

Analysis

The measurement projects the state of Alice, suppose for the analysis that she get $|B_{01}\rangle$ (the same argument can be done for any other bell states), what this implies is that our state is projected onto $|B_{01}\rangle$ using the following projector

$$|B_{01}\rangle_{12}\langle B_{01}|_{12}$$

In Bob's lab nothing has happened which implies that the first operation is a construction using the above projector and the identity applied to our state which gives the following

$$(|B_{01}\rangle_{12}\langle B_{01}|_{12} \otimes I_3)(|\varphi\rangle_1 \otimes |B_{00}\rangle_{23}) = |B_{01}\rangle_{12} \otimes \underbrace{\langle B_{01}|_{12}|\varphi\rangle_1 \otimes |B_{00}\rangle_{23}}_{\text{state vector for 3 in Bob's lab}}$$

As we can see, there is not entanglement now between 2 and 3 but 1 and 2 are now entangled. Let us now compute the right par of the RHS which gives us

$$\begin{aligned}&\frac{1}{\sqrt{2}}(\langle 0|_1 \otimes \langle 1|_2 + \langle 1|_1 \otimes \langle 0|_1)(\alpha|0\rangle_1 + \beta|1\rangle_1) \otimes \frac{1}{\sqrt{2}}(|0\rangle_2 \otimes |0\rangle_3 + |1\rangle_2 \otimes |1\rangle_3)\\ &= \frac{1}{\sqrt{2}}(\alpha\langle 1|_2 + \beta\langle 0|_2) \otimes \frac{1}{\sqrt{2}}(|0\rangle_2 \otimes |0\rangle_3 + |1\rangle_2 \otimes |1\rangle_3)\\ &= \frac{1}{2}(\alpha|1\rangle_3 + \beta|0\rangle_3)\end{aligned}$$

But as we didn't normalized after the measurement we should do it now giving us

$$= \alpha|1\rangle_3 + \beta|0\rangle_3$$

Then after this he receives from Alice the two bits 01 which means that he needs to apply X_3 giving us

$$\begin{aligned}X(\alpha|1\rangle_3 + \beta|0\rangle_3) &= \alpha X|1\rangle + \beta X|0\rangle\\ &= \alpha|0\rangle + \beta|1\rangle = |\varphi\rangle_3\end{aligned}$$

5.1.3 Dense coding

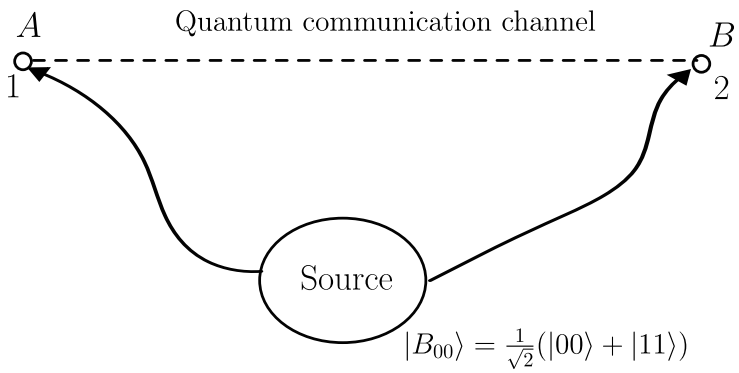


Figure 5.1: Dense coding protocol

The goal is that Alice transmits two classical bits to Bob using only one qbit. A source prepare an EPR pair in state $|B_{00}\rangle$ with one particle sent to Alice and the other to Bob.

Protocol

If Alice wants to send 00, A will sends with I_1 and send the qbit to Bob $\implies$ Bob receives full EPR pair $|B_{00}\rangle_{\text{Bob}}$

- If Alice want to send 01, Alice will apply X_1 and send the qbit to Bob $\implies$ Bob receives $(X_1 \otimes I_2)|B_{00}\rangle = |B_{01}\rangle_{\text{Bob}}$
- If Alice want to send 10, Alice will apply Z_1 and send the qbit to Bob $\implies$ Bob receives $(Z_1 \otimes I_2)|B_{00}\rangle = |B_{10}\rangle_{\text{Bob}}$
- If Alice want to send 11, Alice will apply Z_1X_1 and send the qbit to Bob $\implies$ Bob receives $(Z_1X_1 \otimes I_2)|B_{00}\rangle = |B_{11}\rangle_{\text{Bob}}$

Finally, Bob Measure the full pair in the Bell basis which the output will be $|B_{ij}\rangle_{\text{received}}$ Where ij is the bits that Alice sent.

Let us work through the example when Alice wants to send 01. In that case which the state that Bob will receive is

$$\begin{aligned}(X_1 \otimes I_2)\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) &= ((|0\rangle\langle 1| + |1\rangle\langle 0|) \otimes (|0\rangle\langle 0| + |1\rangle\langle 1|))\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)\\ &= \frac{1}{\sqrt{2}}(|1\rangle \otimes |0\rangle + |0\rangle \otimes |1\rangle) = |B_{01}\rangle\end{aligned}$$

Now one could do it for all but I am too lazy!

5.2 Bell Inequalities

In this section we will cover only one inequality, the CHSH inequality which was first proposed in 1969. Let us first start with the experimental setting :

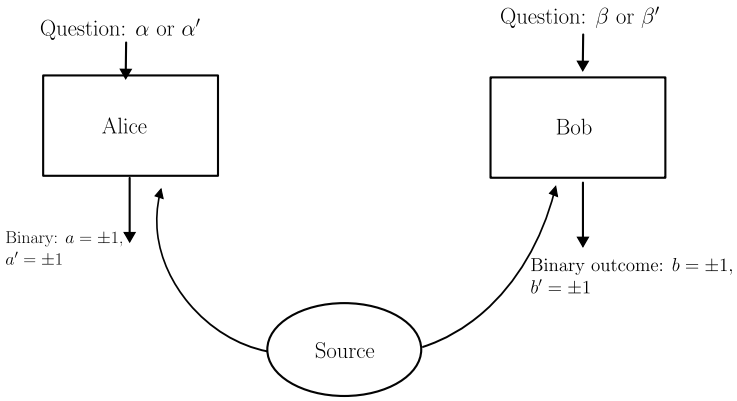


Figure 5.2: Experimental Setting

We have a source giving state to both Alice and Bob, the source produce a pair of photons. One to Alice and the other to Bob. Each Alice and Bob can choose with each of the following basis they can make a measurement with :

$$\begin{cases} |\alpha\rangle = \cos\alpha|0\rangle + \sin\alpha|1\rangle \\ |\alpha_\perp\rangle = -\sin\alpha|0\rangle + \cos\alpha|1\rangle \end{cases}$$
$$\left\{|\alpha'\rangle, |\alpha'_\perp\rangle\right.$$

The outcome of those measurements is given as random variables

$$a = \pm 1$$
$$a' = \pm 1$$

Bob also measures with a basis

$$\{|\beta\rangle, |\beta_\perp\rangle\}$$
$$\{|\beta'\rangle, |\beta'_\perp\rangle\}$$

With the outcome of the measurements being also random variables

$$b = \pm 1$$
$$b' = \pm 1$$

Which gives four possibilities

$$(\alpha, \beta) \to (a, b)$$
$$(\alpha', \beta) \to (a', b)$$
$$(\alpha, \beta') \to (a, b')$$
$$(\alpha', \beta') \to (a', b')$$

Let us also define the four possible experimental configurations

$$1 = (\alpha, \beta), 2 = (\alpha, \beta'), 3 = (\alpha', \beta), 4 = (\alpha', \beta')$$

What we will be doing in this experiment it to have a lot of '*trials*' meaning we ask, for n measurement we seek to do some sort of statistical analysis on the outcome of the experiment in order to deduce some kind of probability distribution.

After having measured the outcome, Alice and Bob communicate and exchange their results. After having collected everything, they will compute the **correlation coefficient** which is given as (knowing $N_1 \approx N_2 \approx N_3 \approx N_4 \approx \frac{N}{4}$ which are the number of event in each experimental configurations) :

$$X^{\text{experimental}} = \frac{1}{N_1} \sum_{i_1:(\alpha,\beta)} a_{i_1} b_{i_1} + \frac{1}{N_2} \sum_{i_2:(\alpha,\beta')} a_{i_2} b'_{i_2} - \frac{1}{N_3} \sum_{i_3:(\alpha',\beta)} a'_{i_3} b_{i_3} + \frac{1}{N_4} \sum_{i_4:(\alpha',\beta')} a'_{i_4} b'_{i_4}$$

Theoretical prediction

Now assume that we are in the case were we don't believe in the true randomness of quantum, meaning that we believe that if we were to know all that there is to know, The theoretical prediction of our experiment is given as

$$X^{\text{theory}} = \mathbb{E}_{\alpha\beta}(ab) + \mathbb{E}_{\alpha\beta'}(ab') - \mathbb{E}_{\alpha'\beta}(a'b) + \mathbb{E}_{\alpha'\beta'}(a'b')$$

To be able to compute this we need to introduce the hidden variable local theory in order to give a view point of a '*classical physicist*'.

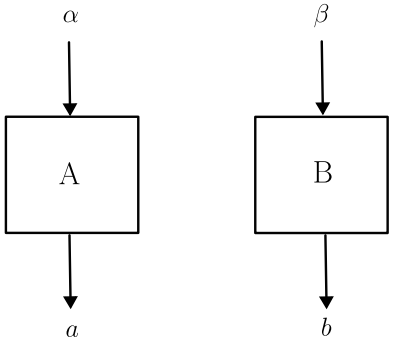
What we are saying is that the outcome of a measurement of α into a is given by the probability.

$$P_A(a|\alpha, \lambda)$$

Where λ is a collection of hidden variables, meaning that the outcome distribution is determined by λ and the setting α. What's important here is also the '*local*' word, the result of a does not depend on β, the experiment of Alice is not influenced by any other factor.

The same mechanism works with Bob giving

$$P_B(b|\beta, \lambda)$$



Which can be formalized as

$$P(a, b|\alpha, \beta, \lambda) = P_A(a|\alpha, \lambda) P_B(b|\beta, \lambda)$$

We also assume that the hidden variables are distributed according to some distribution $h(\lambda)$ which does **not** depend on the settings α, β (**measurement independence** or '*free choice*'). This is the third standard assumption alongside locality and realism (hidden variables): without it, we could not use the same $h(\lambda)$ in all four expectations.

Theorem 3. CHSH inequality (Bell inequality)

$$-2 \leq X^{\text{LHV theory}} \leq 2$$

Proof. *Let us first try to compute $\mathbb{E}_{\alpha,\beta}(ab)$. To do so we first need to compute the conditional over λ times the probability of λ which implies integrating over λ:*

$$\begin{aligned}\mathbb{E}_{\alpha,\beta}(ab) &= \sum_{a,b=\pm 1} ab \int d\lambda P(a,b|\alpha,\beta,\lambda)h(\lambda) \\ &= \sum_{a,b=\pm 1} ab \int d\lambda P_A(a|\alpha,\lambda)P_B(b|\beta,\lambda)h(\lambda)\end{aligned}$$

Note that $h(\lambda) \geq 0$ and $\int h(\lambda)d\lambda = 1$, also let's add (magically) the probability distribution of a' and b' (which shouldn't change anything as their sums are 1).

$$\begin{aligned}&= \sum_{a,b} ab \int d\lambda P_A(a|\alpha,\lambda)P_B(b|\beta,\lambda)\left[\sum_{a'} P_A(a'|\alpha',\lambda)\right]\left[\sum_{b'} P_B(b'|\beta',\lambda)\right] \\ &= \sum_{a,a',b,b'\in\{\pm 1\}} ab \int d\lambda P_A(a'|\alpha',\lambda)P_A(a|\alpha,\lambda)P_B(b'|\beta',\lambda)P_B(b|\beta,\lambda)h(\lambda)\end{aligned}$$

Let us now define

$$Q(a,a',b,b'|\alpha,\alpha',\beta,\beta') = \int d\lambda P_A(a|\alpha,\lambda)P_A(a'|\alpha',\lambda)P_B(b|\beta,\lambda)P_B(b'|\beta',\lambda)h(\lambda)$$

Let us simplify the notation by removing the conditional for convenience (they are still there), which implies that our expected value $\mathbb{E}_{\alpha\beta}(ab)$ can be rewritten as

$$\mathbb{E}_{\alpha,\beta}(ab) = \sum_{a,a',b,b'} ab\, Q(a,a',b,b')$$

The key idea is that Q is a genuine joint probability distribution over all four outcomes: $Q \geq 0$ and $\sum_{a,a',b,b'} Q(a,a',b,b') = 1$. Locality plus hidden variables guarantees that such a joint distribution exists, even though a and a' (or b and b') can never be measured together. This is the whole content of the theorem. Which means that our random variable can be rewritten as

$$\begin{aligned}X^{LHV\ theory} &= \sum_{a,a',b,b'} Q(a,a',b,b')[ab + ab' - a'b + a'b'] \\ &= \sum_{a,a',b,b'} Q(a,a',b,b')[a(b+b') + a'(b'-b)]\end{aligned}$$

Notice here that all random variables here are binary which means that $(b+b')$ can either be $0, 2$ or -2. Notice now that the case $b = b'$ gives $2ab + 0 = \pm 2$, and if $b \neq b'$ (e.g. $b = 1, b' = -1$) we get $0 - 2a' = \pm 2$. We can convince ourselves that the result of the $[\cdot] = \pm 2$. Therefore

$$\left|X^{LHV\ theory}\right| \leq \sum_{a,a',b,b'} Q(a,a',b,b')|[\cdot]| = 2\sum_{a,a',b,b'} Q(a,a',b,b') = 2$$

□

We can show that the hidden variable local theory cannot be applied to quantum mechanics! Meaning that the joint distribution of our two states here cannot be factorized into a product of two probability distributions. Intuitively, this is related to Bell states or more generally entangled states (which cannot be factorized into tensor products). Concretely, we have that for EPR Bell pairs distributed by some

$$\lambda \leftrightarrow |B_{00}\rangle = \frac{1}{\sqrt{2}}(|0_A 0_B\rangle + |1_A 1_B\rangle)$$

We find that

$$\max_{\alpha,\beta,\alpha',\beta'} X^{\text{QM theory}}(\alpha,\beta,\alpha',\beta') = 2\sqrt{2}$$

Which therefore implies that quantum mechanics cannot be represented using the hidden variable local theory, meaning technically that quantum mechanics is **non-local**.

Quantum prediction for Bell pairs (EPR pairs)

Let us now look at the same '*experiment*' but with Bell pairs, we have a source of qubits for $i = 1, \dots, N$ with a distribution $|B_{00}\rangle = \frac{1}{\sqrt{2}}(|0\rangle_A \otimes |0\rangle_B + |1\rangle_A \otimes |1\rangle_B)$ with A and B observables defined as

$$A = (+1)|\alpha\rangle\langle\alpha| + (-1)|\alpha_\perp\rangle\langle\alpha_\perp|$$
$$A' = (+1)|\alpha'\rangle\langle\alpha'| + (-1)|\alpha'_\perp\rangle\langle\alpha'_\perp|$$

And the same for Bob

$$B = (+1)|\beta\rangle\langle\beta| + (-1)|\beta_\perp\rangle\langle\beta_\perp|$$
$$B' = (+1)|\beta'\rangle\langle\beta'| + (-1)|\beta'_\perp\rangle\langle\beta'_\perp|$$

Our goal is to compute the expected value as in the classical case which is given as

$$\begin{aligned} X^{\text{QM theory}} &= \langle B_{00}|A \otimes B|B_{00}\rangle + \langle B_{00}|A \otimes B'|B_{00}\rangle - \langle B_{00}|A' \otimes B|B_{00}\rangle + \langle B_{00}|A' \otimes B'|B_{00}\rangle \\ &= \cos(2(\alpha-\beta)) + \cos(2(\alpha-\beta')) - \cos(2(\alpha'-\beta)) + \cos(2(\alpha'-\beta')) \end{aligned}$$

The proof is left as an exercise. Then what is left to do is to see if our $X^{\text{QM theory}}$ is contained in the interval that we have told before ($\left[-2\sqrt{2}, 2\sqrt{2}\right]$). If we maximize our function with respect to $\alpha, \alpha', \beta, \beta'$ what we get is

$$\alpha = 0 \qquad \alpha' = -\frac{\pi}{4}$$
$$\beta = \frac{\pi}{8} \qquad \beta' = -\frac{\pi}{8}$$

Which gives

$$\max X^{\text{QM theory}} = 2\sqrt{2}$$

Okay so we see that formalizing as before doesn't work as well, a question that is arising is the following :

$$P(a, b|\alpha, \beta, \text{given Bell states}) =?$$

Answer

$$= \frac{1}{4}(1 + ab\cos(2(\alpha-\beta))) \neq f(a|\alpha)f(b|\beta)$$

Which implies that the following distribution **cannot** be factorized. We therefore say that quantum mechanics is **non-local**.

Development

Let us develop here what we have first:

$$\begin{aligned} A \otimes B|B_{00}\rangle &= (|\alpha\rangle\langle\alpha| - |\alpha_\perp\rangle\langle\alpha_\perp|) \otimes (|\beta\rangle\langle\beta| - |\beta_\perp\rangle\langle\beta_\perp|)\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \\ &= ((\cos\alpha|0\rangle + \sin\alpha|1\rangle)(\cos\alpha\langle 0| + \sin\alpha\langle 1|) - (-\sin\alpha|0\rangle + \cos\alpha|1\rangle)(-\sin\alpha\langle 0| + \cos\alpha\langle 1|)) \\ &\quad \otimes ((\cos\beta|0\rangle + \sin\beta|1\rangle)(\cos\beta\langle 0| + \sin\beta\langle 1|) - (-\sin\beta|0\rangle + \cos\beta|1\rangle)(-\sin\beta\langle 0| + \cos\beta\langle 1|))\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \\ &= \big(\cos^2\alpha|0\rangle\langle 0| + \cos\alpha\sin\alpha(|0\rangle\langle 1| + |1\rangle\langle 0|) + \sin^2\alpha|1\rangle\langle 1|\big) \\ &\quad - \big(\sin^2\alpha|0\rangle\langle 0| - \sin\alpha\cos\alpha(|0\rangle\langle 1| + |1\rangle\langle 0|) + \cos^2\alpha|1\rangle\langle 1|\big) \\ &\quad \otimes \big(\cos^2\beta|0\rangle\langle 0| + \cos\beta\sin\beta(|0\rangle\langle 1| + |1\rangle\langle 0|) + \sin^2\beta|1\rangle\langle 1|\big) \\ &\quad - \big(\sin^2\beta|0\rangle\langle 0| - \sin\beta\cos\beta(|0\rangle\langle 1| + |1\rangle\langle 0|) + \cos^2\beta|1\rangle\langle 1|\big)|B_{00}\rangle \\ &= \big(\cos^2\alpha + \cos\alpha\sin\alpha - \sin^2\alpha - \sin\alpha\cos\alpha\big)|0\rangle + \big(\sin^2\alpha + \cos\alpha\sin\alpha - \sin\alpha\cos\alpha - \cos^2\alpha\big)|1\rangle \\ &= \big(\cos^2\alpha - \sin^2\alpha\big)|0\rangle + \big(\sin^2\alpha - \cos^2\alpha\big)|1\rangle \end{aligned}$$

Now applying the inner product with $\langle B_{00}|$ gives

$$\frac{1}{\sqrt{2}}(\langle 00| + \langle 11|)\big(\big(\cos^2\alpha - \sin^2\alpha\big)|0\rangle + \big(\sin^2\alpha - \cos^2\alpha\big)|1\rangle\big) =$$

Suppose that the source distributes a state $|\varphi_A\rangle \otimes |\varphi_B\rangle$, we ask ourselves what should be $X^{\text{QM, tensor prod}}$? Can this be bounded by ± 2? (exercise)

5.3 Ekert (1991) QKD protocol

The goal of this protocol is also to generate a shared key for the one-time pad. This time we use entanglement: a Bell test lets Alice and Bob detect an eavesdropper and bound the information she can have about the key.

Setup

Alice and Bob share, at each time instant, a Bell pair in state $|B_{00}\rangle$. Measuring along angle θ means using the basis $\{|\theta\rangle, |\theta_\perp\rangle\}$ with $|\theta\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle$. For $i = 1, \ldots, N$:

- Alice measures her qubit along an angle chosen at random from $\alpha_1 = -\frac{\pi}{4}, \alpha_2 = -\frac{\pi}{8}, \alpha_3 = 0$.
- Bob measures his qubit along an angle chosen at random from $\beta_1 = -\frac{\pi}{8}, \beta_2 = 0, \beta_3 = \frac{\pi}{8}$.
- Alice records $x_i = \pm 1$ according to the outcome α or $\alpha_\perp$.
- Bob records $y_i = \pm 1$ according to the outcome β or $\beta_\perp$.

All of this is done without communication. For $|B_{00}\rangle$,

$$P(x_i = y_i) = \cos^2(\alpha - \beta), \qquad E(\alpha, \beta) = \langle x_i y_i \rangle = \cos 2(\alpha - \beta).$$

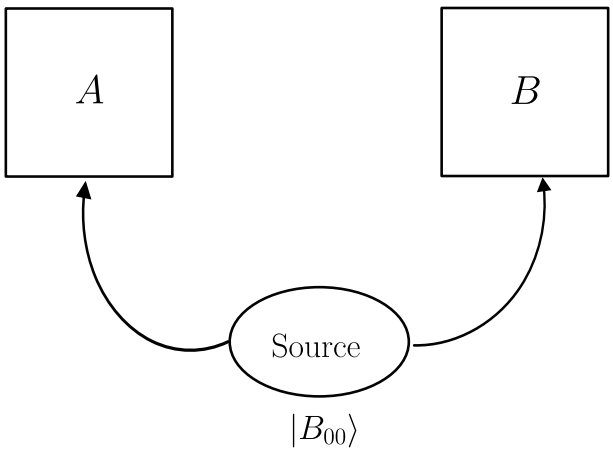


Figure 5.3: Ekert protocol

Public communication phase

After all qubits have been measured, Alice and Bob publicly announce their basis choices for each $i = 1, \ldots, N$, and sort the rounds:

- **Test rounds**: $(\alpha_3, \beta_3), (\alpha_3, \beta_1), (\alpha_1, \beta_1), (\alpha_1, \beta_3)$. They publicly reveal their outcomes for these rounds and estimate

$$S = E(\alpha_3, \beta_3) + E(\alpha_3, \beta_1) + E(\alpha_1, \beta_1) - E(\alpha_1, \beta_3).$$

  Each correlator is only $\pm\frac{1}{\sqrt{2}}$, but the combination gives $S = 2\sqrt{2}$, the maximal quantum value. If $S \leq 2$, the correlations could be classical (pre-determined values Eve could copy): abort. If S is close to $2\sqrt{2}$, the pairs are close to maximally entangled, and by monogamy of entanglement Eve's information is small. These rounds are then discarded, since their outcomes are public.
- **Key rounds**: (α_3, β_2) and (α_2, β_1). Their outcomes are never revealed.
- The remaining three combinations are discarded.

Secret key generation

In the key rounds the settings are

$$\begin{aligned} (\alpha_3, \beta_2) \quad &\text{or} \quad (\alpha_2, \beta_1) \\ (0, 0) \quad &\text{or} \quad \left(-\frac{\pi}{8}, -\frac{\pi}{8}\right) \end{aligned}$$

Since the angles are equal, $P(x_i = y_i) = \cos^2(0) = 1$: ideally $x_i = y_i$. Mapping $\pm 1 \mapsto 0, 1$ gives the shared key. The values of these bits are never made public. In practice, noise (and Eve) cause a few mismatches, so the key is finished with error correction and privacy amplification, which shortens it according to how far S is below $2\sqrt{2}$.

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